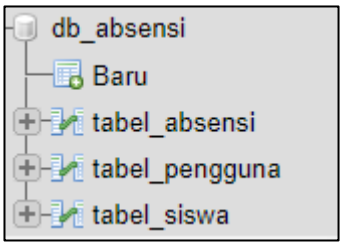
Nama:

Struktur Database: db_absensi



tabel_absensi

#	Nama	Jenis	Penyortiran	Atribut	Tak Ternilai	Bawaan	Komentar	Ekstra
1	idabsensi	int(11)			Tidak	*Tidak ada*		AUTO_INCREMENT
2	nis	varchar(50)	utf8mb4_general_ci		Tidak	*Tidak ada*		
3	tanggal	date			Tidak	*Tidak ada*		
4	jam_masuk	time			Tidak	*Tidak ada*		
5	jam_pulang	time			Tidak	*Tidak ada*		
6	keterangan	varchar(10)	utf8mb4_general_ci		Tidak	*Tidak ada*		

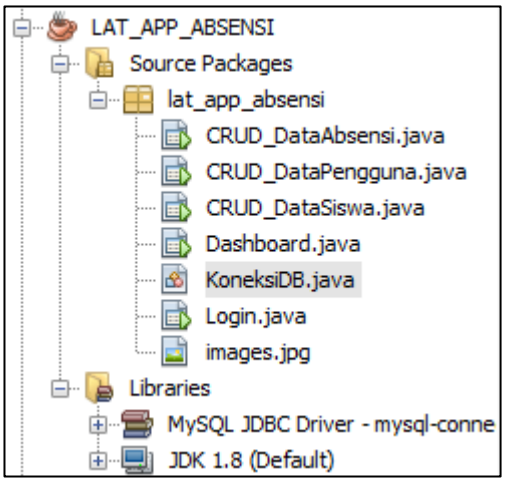
tabel_pengguna

#	Nama	Jenis	Penyortiran	Atribut	Tak Ternilai	Bawaan	Komentar	Ekstra
1	idpengguna	int(11)			Tidak	*Tidak ada*		AUTO_INCREMENT
2	username	varchar(100)	utf8mb4_general_ci		Tidak	*Tidak ada*		
3	password	varchar(100)	utf8mb4_general_ci		Tidak	*Tidak ada*		
4	level	varchar(50)	utf8mb4_general_ci		Tidak	*Tidak ada*		

tabel_siswa

#	Nama	Jenis	Penyortiran	Atribut	Tak Ternilai	Bawaan	Komentar	Ekstra
1	idsiswa	int(11)			Tidak	*Tidak ada*		AUTO_INCREMENT
2	nis	varchar(50)	utf8mb4_general_ci		Tidak	*Tidak ada*		
3	nama	varchar(100)	utf8mb4_general_ci		Tidak	*Tidak ada*		
4	kelas	varchar(30)	utf8mb4_general_ci		Tidak	*Tidak ada*		

Struktur Direktori



KoneksiDB.java

```
package lat_app_absensi;

import java.sql.*;
import javax.swing.JOptionPane;


public class KoneksiDB {

   static Connection con;

   public static Connection getConnection() {
        try {
           con = DriverManager.getConnection("jdbc:mysql://localhost:3306/db_absensi", "root", "");
        } catch (Exception e) {
           JOptionPane.showMessageDialog(null, "Error : Koneksi Gagal!");
           System.exit(0);
        }
        return con;
   }
}
```

Tambahan di Dashboard.java

```
private void buttonLogoutActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      this.dispose();
      new Login().setVisible(true);
   }

   private void buttonDataSiswaActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      this.dispose();
      new CRUD_DataSiswa().setVisible(true);
   }

   private void buttonDataPenggunaActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      this.dispose();
      new CRUD_DataPengguna().setVisible(true);
   }

   private void buttonDataAbsensiActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      this.dispose();
      new CRUD_DataAbsensi().setVisible(true);
   }
```

CRUD_DataAbsensi.java

```
package lat_app_absensi;

import java.beans.Statement;
import java.sql.Connection;
import java.sql.ResultSet;
import java.time.LocalDateTime;
import java.time.format.DateTimeFormatter;
import javax.swing.DefaultComboBoxModel;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;

public class CRUD_DataAbsensi extends javax.swing.JFrame {

    Statement st;
    Connection con = KoneksiDB.getConnection();
    ResultSet rs;
    DefaultComboBoxModel modelc;
    DefaultTableModel model;
    LocalDateTime myDateObj = LocalDateTime.now();

    /**
     * Creates new form CRUD_DataAbsensi
     */
    public CRUD_DataAbsensi() {
        initComponents();
        awal();
    }

    private void awal() {
        tabelAbsensi();
        jID.setEnabled(false);
        jID.setText("Auto Fill");
        jNIS.setText("");
        jTanggal.setText(myDateObj.format(DateTimeFormatter.ISO_DATE));
        jMasuk.setText(myDateObj.format(DateTimeFormatter.ISO_TIME).substring(0, 8));
        jPulang.setText(myDateObj.format(DateTimeFormatter.ISO_TIME).substring(0, 8));
        jNama.setEnabled(false);
        jNama.setText("Based on NIS");
        buttonUbah.setEnabled(false);
        buttonHapus.setEnabled(false);
        buttonTambah.setEnabled(true);
    }

    private void tabelAbsensi() {
        String[] judul = {"ID", "NIS", "Nama", "Tanggal", "Jam Masuk", "Jam Pulang", "Keterangan"};
        model = new DefaultTableModel(judul, 0);
        tabelAbsensi.setModel(model);
        String sql = "SELECT a.*, b.nama FROM tabel_absensi a "
                + "INNER JOIN tabel_siswa b ON a.nis = b.nis";

        try {
            rs = con.createStatement().executeQuery(sql);
            while (rs.next()) {
                String id = rs.getString("idabsensi");
                String nis = rs.getString("nis");
                String nama = rs.getString("nama");
                String tanggal = rs.getString("tanggal");
                String masuk = rs.getString("jam_masuk");
                String pulang = rs.getString("jam_pulang");
                String keterangan = rs.getString("keterangan");

                String[] data = {id, nis, nama, tanggal, masuk, pulang, keterangan};
                model.addRow(data);
            }
        } catch (Exception e) {
            System.out.println(e);
        }
    }
```

```
private void buttonDashboardActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    this.dispose();
    new Dashboard().setVisible(true);
  }

private void tabelAbsensiMouseClicked(java.awt.event.MouseEvent evt) {
    // TODO add your handling code here:
    int i = tabelAbsensi.getSelectedRow();

    if (i > -1) {
      jID.setText(model.getValueAt(i, 0).toString());
      jNIS.setText(model.getValueAt(i, 1).toString());
      jNama.setText(model.getValueAt(i, 2).toString());
      jTanggal.setText(model.getValueAt(i, 3).toString());
      jMasuk.setText(model.getValueAt(i, 4).toString());
      jPulang.setText(model.getValueAt(i, 5).toString());
      jKeterangan.setSelectedItem(model.getValueAt(i, 6));
    }

    //disable item on klik
    buttonTambah.setEnabled(false);
    buttonUbah.setEnabled(true);
    buttonHapus.setEnabled(true);
  }

  private void jNISKeyReleased(java.awt.event.KeyEvent evt) {
    // TODO add your handling code here:
    try {
      String sql = "select nama from tabel_siswa WHERE nis='" + jNIS.getText() + "'";
      rs = con.createStatement().executeQuery(sql);
      while (rs.next()) {
        jNama.setText(rs.getString("nama"));
      }
    } catch (Exception e) {
      System.out.println(e.getMessage());
    }
  }

  private void buttonHapusActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    try {
      int confirm = JOptionPane.showConfirmDialog(this,
          "Apakah Anda yakin ingin menghapus data absensi " + jNIS.getText() + "?",
          "Konfirmasi", JOptionPane.YES_NO_OPTION);
      if (confirm == JOptionPane.YES_OPTION) {
        String sql = "DELETE from tabel_absensi WHERE idabsensi = " + jID.getText() + "";
        con.createStatement().executeUpdate(sql);
        JOptionPane.showMessageDialog(null, "Berhasil Dihapus!");
        awal();
      } else if (confirm == JOptionPane.NO_OPTION) {
        //
      }
    } catch (Exception e) {
      JOptionPane.showMessageDialog(null, "Gagal Dihapus!");
    }
  }
```

```
private void buttonTambahActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    if (!jNIS.getText().isEmpty()) {
      try {
        String sql = "INSERT INTO tabel_absensi value (null,"
            + "'" + jNIS.getText() + "',"
            + "'" + jTanggal.getText() + "',"
            + "'" + jMasuk.getText() + "',"
            + "'" + jPulang.getText() + "',"
            + "'" + jKeterangan.getSelectedItem() + "'"
            + ")";
        con.createStatement().executeUpdate(sql);
        JOptionPane.showMessageDialog(null, "Berhasil Tambah Data!");
        awal();
      } catch (Exception e) {
        JOptionPane.showMessageDialog(null, "Gagal Tambah Data!");
      }
    } else {
      if (jNIS.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "NIS wajib diisi");
      }
    }
  }

private void jKeteranganItemStateChanged(java.awt.event.ItemEvent evt) {
    // TODO add your handling code here:
    if (jKeterangan.getSelectedItem().equals("Masuk")) {
      jMasuk.setText(myDateObj.format(DateTimeFormatter.ISO_TIME).substring(0, 8));
      jPulang.setText(myDateObj.format(DateTimeFormatter.ISO_TIME).substring(0, 8));
    } else {
      jMasuk.setText("00:00:00");
      jPulang.setText("00:00:00");
    }
  }

  private void buttonUbahActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    if (!jNIS.getText().isEmpty()) {
      try {
        String sql = "UPDATE tabel_absensi SET"
            + " nis ='" + jNIS.getText() + "',"
            + " tanggal ='" + jTanggal.getText() + "',"
            + " jam_masuk ='" + jMasuk.getText() + "',"
            + " jam_pulang ='" + jPulang.getText() + "',"
            + " keterangan ='" + jKeterangan.getSelectedItem() + "'"
            + " WHERE idabsensi = " + jID.getText() + "";
        con.createStatement().executeUpdate(sql);
        JOptionPane.showMessageDialog(null, "Berhasil Diubah!");
        awal();
      } catch (Exception e) {
        JOptionPane.showMessageDialog(null, "Gagal Diubah!");
      }
    } else {
      if (jNIS.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "NIS wajib diisi");
      }
    }
  }
```

CRUD_DataPengguna.java

```
package lat_app_absensi;

import java.beans.Statement;
import java.sql.Connection;
import java.sql.ResultSet;
import javax.swing.DefaultComboBoxModel;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;

public class CRUD_DataPengguna extends javax.swing.JFrame {

    Statement st;
    Connection con = KoneksiDB.getConnection();
    ResultSet rs;
    DefaultComboBoxModel modelc;
    DefaultTableModel model;

    /**
     * Creates new form CRUD_DataPengguna
     */
    public CRUD_DataPengguna() {
        initComponents();
        awal();
    }

    private void awal() {
        tabelPengguna();
        jID.setEnabled(false);
        jID.setText("Auto Fill");
        jUsername.setText("");
        jPassword.setText("");
        jRePassword.setText("");
        buttonTambah.setEnabled(true);
        buttonUbah.setEnabled(false);
        buttonHapus.setEnabled(false);
    }

    private void tabelPengguna() {
        String[] judul = {"ID", "Username", "Password", "Level"};
        model = new DefaultTableModel(judul, 0);
        tabelPengguna.setModel(model);
        String sql = "SELECT * FROM tabel_pengguna";

        try {
            rs = con.createStatement().executeQuery(sql);
            while (rs.next()) {
                String id = rs.getString("idpengguna");
                String username = rs.getString("username");
                String password = rs.getString("password");
                String level = rs.getString("level");

                String[] data = {id, username, password, level};
                model.addRow(data);
            }
        } catch (Exception e) {
            System.out.println(e);
        }
    }
```

```
private void buttonDashboardActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    this.dispose();
    new Dashboard().setVisible(true);
  }

  private void tabelPenggunaMouseClicked(java.awt.event.MouseEvent evt) {
    // TODO add your handling code here:
    int i = tabelPengguna.getSelectedRow();

    if (i > -1) {
      jID.setText(model.getValueAt(i, 0).toString());
      jUsername.setText(model.getValueAt(i, 1).toString());
      jPassword.setText(model.getValueAt(i, 2).toString());
      jRePassword.setText(model.getValueAt(i, 2).toString());
      jLevel.setSelectedItem(model.getValueAt(i, 3));
    }

    //disable item on klik
    buttonTambah.setEnabled(false);
    buttonUbah.setEnabled(true);
    buttonHapus.setEnabled(true);
  }

  private void buttonHapusActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    try {
      int confirm = JOptionPane.showConfirmDialog(this,
          "Apakah Anda yakin ingin menghapus data dengan username " + jUsername.getText() + "?",
          "Konfirmasi", JOptionPane.YES_NO_OPTION);
      if (confirm == JOptionPane.YES_OPTION) {
        String sql = "DELETE from tabel_pengguna WHERE idpengguna = " + jID.getText() + "";
        con.createStatement().executeUpdate(sql);
        JOptionPane.showMessageDialog(null, "Berhasil Dihapus!");
        awal();
      } else if (confirm == JOptionPane.NO_OPTION) {
        //
      }
    } catch (Exception e) {
      JOptionPane.showMessageDialog(null, "Gagal Dihapus!");
    }
  }
```

```
private void buttonUbahActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    if (!jUsername.getText().isEmpty() && !jPassword.getText().isEmpty() && !jRePassword.getText().isEmpty()) {
      // kurang yang password dan repassword
      if (jRePassword.getText().equals(jPassword.getText())) {
        try {
          String sql = "UPDATE tabel_pengguna SET"
              + " username ='" + jUsername.getText() + "',"
              + " password ='" + jPassword.getText() + "',"
              + " level ='" + jLevel.getSelectedItem().toString() + "'"
              + " WHERE idpengguna = " + jID.getText() + "";
          con.createStatement().executeUpdate(sql);
          JOptionPane.showMessageDialog(null, "Berhasil Diubah!");
          awal();
        } catch (Exception e) {
          JOptionPane.showMessageDialog(null, "Gagal Diubah!");
        }
      } else {
        JOptionPane.showMessageDialog(null, "Password dan Retype password Anda tidak sama!");
      }
    } else {
      if (jUsername.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Username wajib diisi");
      }
      if (jPassword.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Password wajib diisi");
      }
      if (jRePassword.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Retype password wajib diisi");
      }
    }
  }

  private void buttonTambahActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    if (!jUsername.getText().isEmpty() && !jPassword.getText().isEmpty() && !jRePassword.getText().isEmpty()) {
      // cek yang password dan repassword
      if (jRePassword.getText().equals(jPassword.getText())) {
        try {
          String sql = "INSERT INTO tabel_pengguna value (null,"
              + "'" + jUsername.getText() + "',"
              + "'" + jPassword.getText() + "',"
              + "'" + jLevel.getSelectedItem().toString() + "'"
              + ")";
          con.createStatement().executeUpdate(sql);
          JOptionPane.showMessageDialog(null, "Berhasil Tambah Data!");
          awal();
        } catch (Exception e) {
          JOptionPane.showMessageDialog(null, "Gagal Tambah Data!");
        }
      } else {
        JOptionPane.showMessageDialog(null, "Password dan Retype password Anda tidak sama!");
      }
    } else {
      if (jUsername.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Username wajib diisi");
      }
      if (jPassword.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Password wajib diisi");
      }
      if (jRePassword.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Retype password wajib diisi");
      }
    }
  }
```

CRUD_DataSiswa.java

```
package lat_app_absensi;

import java.beans.Statement;
import java.sql.Connection;
import java.sql.ResultSet;
import javax.swing.DefaultComboBoxModel;
import javax.swing.JOptionPane;
import javax.swing.table.DefaultTableModel;

public class CRUD_DataSiswa extends javax.swing.JFrame {

    Statement st;
    Connection con = KoneksiDB.getConnection();
    ResultSet rs;
    DefaultComboBoxModel modelc;
    DefaultTableModel model;

    /**
     * Creates new form CRUD_DataSiswa
     */
    public CRUD_DataSiswa() {
        initComponents();
        awal();
    }

    private void awal() {
        tabelSiswa();
        jID.setEnabled(false);
        jID.setText("Auto Fill");
        jNIS.setText("");
        jNama.setText("");
        jKelas.setText("");
        buttonUbah.setEnabled(false);
        buttonHapus.setEnabled(false);
        buttonTambah.setEnabled(true);
    }

    private void tabelSiswa() {
        String[] judul = {"ID", "Nis", "Nama", "Kelas"};
        model = new DefaultTableModel(judul, 0);
        tabelSiswa.setModel(model);
        String sql = "SELECT * FROM tabel_siswa";

        try {
            rs = con.createStatement().executeQuery(sql);
            while (rs.next()) {
                String id = rs.getString("idsiswa");
                String nis = rs.getString("nis");
                String nama = rs.getString("nama");
                String kelas = rs.getString("kelas");

                String[] data = {id, nis, nama, kelas};
                model.addRow(data);
            }
        } catch (Exception e) {
            System.out.println(e);
        }
    }
```

```
private void buttonDashboardActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    this.dispose();
    new Dashboard().setVisible(true);
  }

  private void buttonTambahActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    if (!jNIS.getText().isEmpty() && !jNama.getText().isEmpty() && !jKelas.getText().isEmpty()) {
      try {
        String sql = "INSERT INTO tabel_siswa value (null,"
            + "'" + jNIS.getText() + "',"
            + "'" + jNama.getText() + "',"
            + "'" + jKelas.getText() + "'"
            + ")";
        con.createStatement().executeUpdate(sql);
        JOptionPane.showMessageDialog(null, "Berhasil Tambah Data!");
        awal();
      } catch (Exception e) {
        JOptionPane.showMessageDialog(null, "Gagal Tambah Data!");
      }
    } else {
      if (jNIS.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "NIS wajib diisi");
      }
      if (jNama.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Nama wajib diisi");
      }
      if (jKelas.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Kelas wajib diisi");
      }
    }
  }

  private void tabelSiswaMouseClicked(java.awt.event.MouseEvent evt) {
    // TODO add your handling code here:
    int i = tabelSiswa.getSelectedRow();

    if (i > -1) {
      jID.setText(model.getValueAt(i, 0).toString());
      jNIS.setText(model.getValueAt(i, 1).toString());
      jNama.setText(model.getValueAt(i, 2).toString());
      jKelas.setText(model.getValueAt(i, 3).toString());
    }

    //disable item on klik
    buttonTambah.setEnabled(false);
    buttonUbah.setEnabled(true);
    buttonHapus.setEnabled(true);
  }

  private void buttonHapusActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    try {
      int confirm = JOptionPane.showConfirmDialog(this,
          "Apakah Anda yakin ingin menghapus data dengan NIS=" + jNIS.getText() + "?",
          "Konfirmasi", JOptionPane.YES_NO_OPTION);
      if (confirm == JOptionPane.YES_OPTION) {
        String sql = "DELETE from tabel_siswa WHERE idsiswa = " + jID.getText() + "";
        con.createStatement().executeUpdate(sql);
        JOptionPane.showMessageDialog(null, "Berhasil Dihapus!");
        awal();
      } else if (confirm == JOptionPane.NO_OPTION) {
        //
      }
    } catch (Exception e) {
      JOptionPane.showMessageDialog(null, "Gagal Dihapus!");
    }

  }
```

```
private void buttonUbahActionPerformed(java.awt.event.ActionEvent evt) {
    // TODO add your handling code here:
    if (!jNIS.getText().isEmpty() && !jNama.getText().isEmpty() && !jKelas.getText().isEmpty()) {
      try {
        String sql = "UPDATE tabel_siswa SET"
            + " nis ='" + jNIS.getText() + "',"
            + " nama ='" + jNama.getText() + "',"
            + " kelas ='" + jKelas.getText() + "'"
            + " WHERE idsiswa = " + jID.getText() + "";
        con.createStatement().executeUpdate(sql);
        JOptionPane.showMessageDialog(null, "Berhasil Diubah!");
        awal();
      } catch (Exception e) {
        JOptionPane.showMessageDialog(null, "Gagal Diubah!");
      }
    } else {
      if (jNIS.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "NIS wajib diisi");
      }
      if (jNama.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Nama wajib diisi");
      }
      if (jKelas.getText().isEmpty()) {
        JOptionPane.showMessageDialog(null, "Kelas wajib diisi");
      }
    }
  }
```

Login.java

```
package lat_app_absensi;

import static lat_app_absensi.Dashboard.jPengguna;
import java.beans.Statement;
import java.sql.Connection;
import java.sql.ResultSet;
import javax.swing.JOptionPane;

public class Login extends javax.swing.JFrame {

   Statement st;
   Connection con = KoneksiDB.getConnection();
   ResultSet rs;

   /**
    * Creates new form Login
    */
   public Login() {
      initComponents();
   }

private void buttonCancelActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      System.exit(0);
   }

   private void buttonLoginActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      if (!jUsername.getText().isEmpty() && !jPassword.getText().isEmpty()) {
         try {
            String sql = "SELECT * FROM tabel_pengguna "
                  + "WHERE username = '" + jUsername.getText() + "' and password = '" + jPassword.getText() + "'";
            rs = con.createStatement().executeQuery(sql);
            if (rs.next()) {
               jPengguna.setText("Hi, " + rs.getString("username") + "!");
               this.dispose();
               new Dashboard().setVisible(true);
            } else {
               JOptionPane.showMessageDialog(null, "Username atau password salah");
            }
         } catch (Exception e) {
            System.out.println(e);
         }

      } else {
         JOptionPane.showMessageDialog(null, "Username atau Password wajib diisi");
      }
   }

   private void jPasswordActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      buttonLoginActionPerformed(evt);
   }

   private void jUsernameActionPerformed(java.awt.event.ActionEvent evt) {
      // TODO add your handling code here:
      buttonLoginActionPerformed(evt);
   }
```

PERIKSA KEMBALI DAN SESUAIKAN TANDA PETIK DUA, TANDA KURUNG "()", TANDA KURUNG KURAWAL "{}" PADA SETIAP CLASS MAUPUN METHOD !!